PACER

Psychometric Software

LIVE WEBINAR

Getting Started with PACER

A complete psychometric environment — in your browser, on your desktop, and through Claude

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PACER 3.2.0 · Nexus

MY MISSION

My vision is to build the *leading, industry-grade* psychometric platform.

A comprehensive, modern environment that serves operational testing programs, researchers, and practitioners alike.

🏛️ Operational testing programs · 🔬 Researchers · 👷 Practitioners

— **Harold Doran**, creator of PACER

ONE APPLICATION BRINGS TOGETHER

- ∑ **Advanced psychometric methods**
- ⚡ **High-performance computing**
- ✨ **Emerging AI capabilities**

Making sophisticated analyses easier to **perform**, **integrate**, and **interpret**.

DISCLOSURES

An *independent* project



Independent

PACER is an independent project, not affiliated with or endorsed by any employer or organization.

TM

Trademarks

Claude, Codex, Gemma and other product names are trademarks of their respective owners. Their mention does not imply endorsement.



Views my own

The views expressed in this presentation are my own.

Harold Doran

BOOKMARK THESE

Where to *find everything*

Web app	https://www.paceronline.com
Desktop app New versions and release notes	https://github.com/hdoran/PACER_Download/releases/
Security & Compliance For IT and compliance review	https://www.paceronline.com/security.html
Guided tours	**Take a Tour** button on the landing page · **8 languages**
Template Library	https://www.paceronline.com/templates.html
PACER University	https://www.paceronline.com/university/
Technical Reference	https://www.paceronline.com/pacer-technical-reference.html
What's new	https://www.paceronline.com/changelog.html

THE NEXT 60 MINUTES

Learn the pattern once, *use it everywhere*

0–4
Why PACER
The vision, the links, the problem it solves

4–12
PACER at a glance
Modules, ways to learn, desktop, AI Advisor

12–30
Live demo: the app
IRT calibration and scoring, CDM calibration and profiling

30–48
Live demo: Claude
Optional AI that coordinates a full analysis

48–52
Get started
Your first calibration and next steps

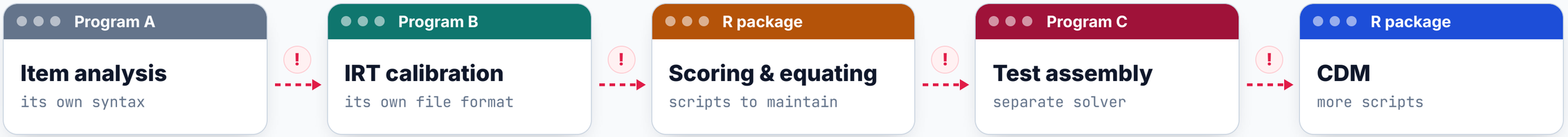
52–60
Q&A
Your questions, your data, your use cases

THE PROBLEM

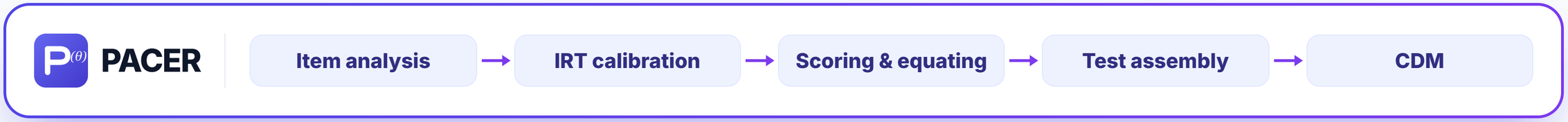
One environment, *not five*

One testing project often runs through **five different programs and R packages**, each with its own syntax, file format and license.

TODAY Every arrow is an export, a reformat and a re-import, and a new place for errors



WITH PACER Same steps, one program: results hand off automatically



✓ One program to install and learn ✓ The same five-step pattern in every module ✓ Optional AI coordinates; PACER does the math

ONE ENVIRONMENT

The full psychometric workflow, *in one place*

Item Analysis
Classical statistics, reliability, and DIF

IRT Calibration
1PL to 3PL, GRM, GPCM, RSM, multigroup, anchoring

IRT Scoring
MLE, MAP, EAP and TCC scoring with diagnostics

Equating
Stocking-Lord, Haebara, Mean/Sigma, Mean/Mean

Test Construction
Automated test assembly to a blueprint

CDM Calibration
G-DINA, A-CDM, LLM, R-RUM, DINA, DINO

Skill Profiler
Skill mastery profiles for every examinee

Latent Regression
Explanatory IRT and plausible values

- Highlighted modules are today's live demos

HELP IS BUILT IN

Six ways to *learn* PACER

Guided tours

Step-by-step walk-throughs; start from **Take a Tour** on the landing page

8 languages

FIRST CLICK

Template Library

Example files in the exact format each module expects

YOUR DATA

PACER University

Interactive lessons on IRT, CDM, equating and more

CONCEPTS

AI Advisor

Ask about the screen you're on; answers come from PACER's own documentation, locally

OPTIONAL · DESKTOP

Claude

Runs and explains whole workflows on your files

OPTIONAL · DESKTOP

Technical Reference

Every model, estimator and algorithm, with the full math

THE MATH

TWO WAYS TO RUN PACER

Same engine, *two front doors*

Web app

https://www.paceronline.com

Nothing to install. Open a browser and start analyzing.

- ✓ Every analysis module
- ✓ Guided tours, templates, University
- ✓ Ideal for trying PACER and for teaching
- – Data is uploaded to the server

BEST FOR a first look, classrooms, quick checks

Desktop app

Windows · macOS · GitHub releases

Recommended

Everything in the web app, running on your own machine. Each release ships with notes on GitHub.

- ✓ Every analysis module
- ✓ Native engine — **fast**
- ✓ Runs **100% air-gapped** — no internet needed
- ✓ Data **never leaves** your computer
- ✓ **Optional** AI Advisor and Claude Desktop

BEST FOR real projects, protected data, AI workflows

WHY A DESKTOP APP

Three reasons to *run it locally*

Speed

The estimation engine runs natively on your machine, in parallel across your processor cores. No uploads, no waiting on a server.

LSAT 2PL, 1,000 × 5 items: 0.01 s estimation

AI, if you want it

A local AI Advisor that knows PACER's documentation, plus a one-click connection that lets Claude run PACER analyses. Both are always optional — off until you turn them on.

Always optional

100% air-gapped

Every analysis runs with no internet connection. Response files are read on your computer and never uploaded — essential for protected student and candidate data.

No network, no upload

OPTIONAL · DESKTOP APP

An AI Advisor *grounded in* PACER

A retrieval-augmented generation (RAG) pipeline built from PACER's own documentation — not a general-purpose chatbot. Like any AI, check what it tells you.

1 · ASK
Your question
Asked from any page; the Advisor knows which module you're on

→

2 · RETRIEVE
PACER knowledge
~1,500 passages from the docs, technical reference and screen guides

→

3 · GENERATE
Gemma 3 (4B)
Google's open-weights model writes the answer from those passages

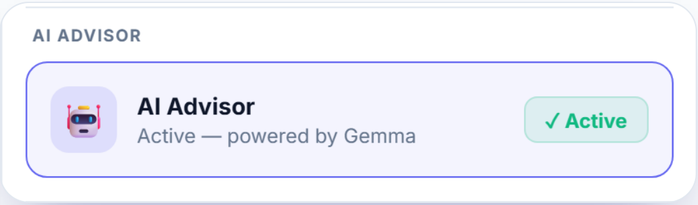
→

4 · ANSWER
PACER-specific help
Options, outputs and next steps for the screen you're on

🧩
Optional
Settings → AI Advisor → Set Up. Skip it and PACER is unchanged.

💻
Installed locally
Runs through Ollama on your computer; one-time ~3 GB download.

🔒
Stays offline
After setup it needs no internet and sends nothing anywhere.



01

PART ONE · LIVE DEMO

The PACER app

📈 IRT Calibration

🎯 IRT Scoring

🧠 CDM Calibration

🗺️ Skill Profiler

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THE ONE THING TO REMEMBER

Every module follows *the same five steps*



STEP 1
Upload
Drop a CSV or Excel file; PACER checks it instantly

STEP 2
Configure
Choose the model and options in the sidebar

STEP 3
Run
One button; results in seconds

STEP 4
Read
Summary cards, tables and interactive plots

STEP 5
Export & hand off
Download CSVs, or send results to the next module

💡 Once you have seen it twice, you know **Item Analysis, Equating and Test Construction** too — they work exactly the same way.

DEMO 1

IRT *calibration and scoring*

IRT Calibration `LSAT.csv · 1,000 × 5`

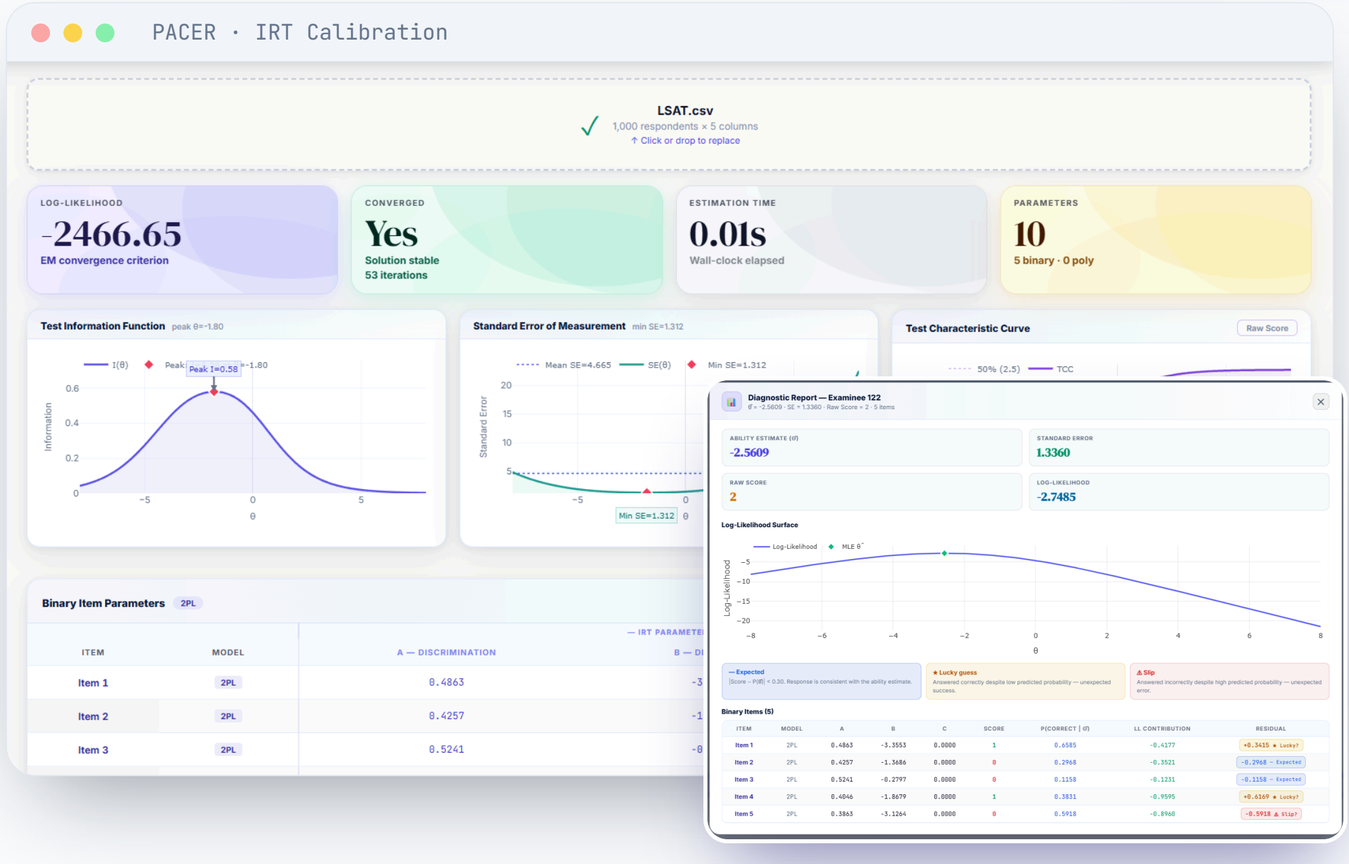
1. Upload the response matrix
2. Choose a 2PL in the sidebar
3. Run; the TIF peaks at θ = −1.8: an easy test
4. Open an item curve and export

IRT Scoring `carries over`

1. Parameters and data load automatically
2. Score with EAP
3. Open one examinee's diagnostic report

REMEMBER THESE FOR LATER

`Item 1: a = 0.4863 · b = -3.3553`



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DEMO 2

CDM calibration and *skill profiles*

gdina_responses.csv + Q-matrix · 10 items · 3 skills · 1,000 examinees

SATURATED

G-DINA

ALL / ANY SKILL

DINA DINO

NO INTERACTIONS · NEW

A-CDM LLM R-RUM

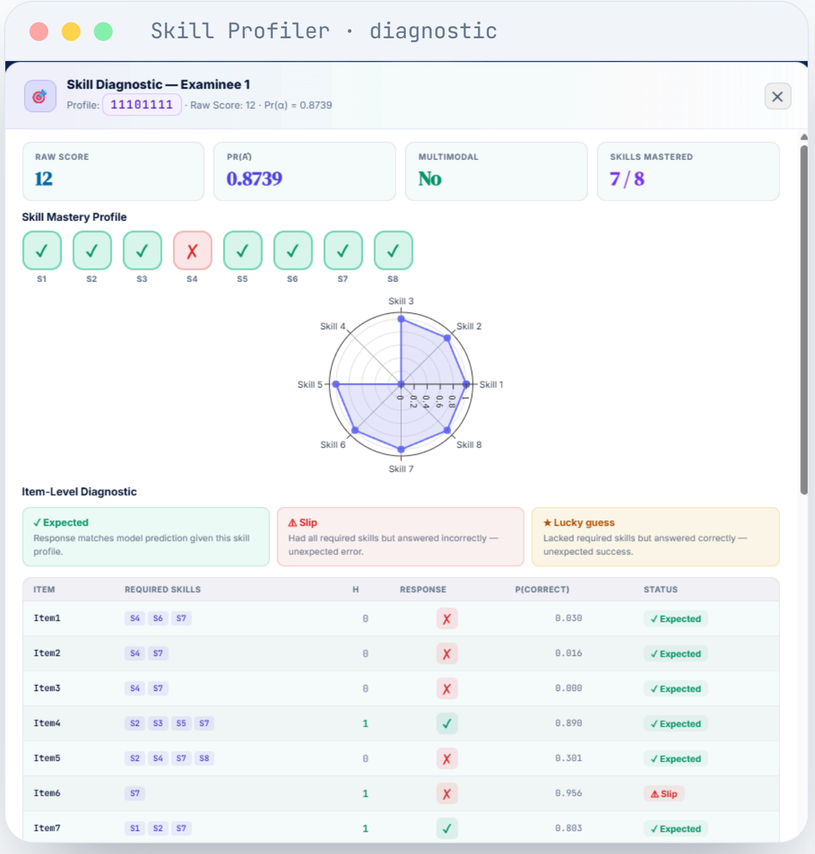
Additive on the probability, logit or log scale

Do we need the most complex model?

BIC · lower is better

R-RUM	12416
G-DINA	12420
LLM	12434
A-CDM	12481
DINA	12507
DINO	12637

A 4-point BIC gap between R-RUM and G-DINA is a near tie: the simpler model competes.



02

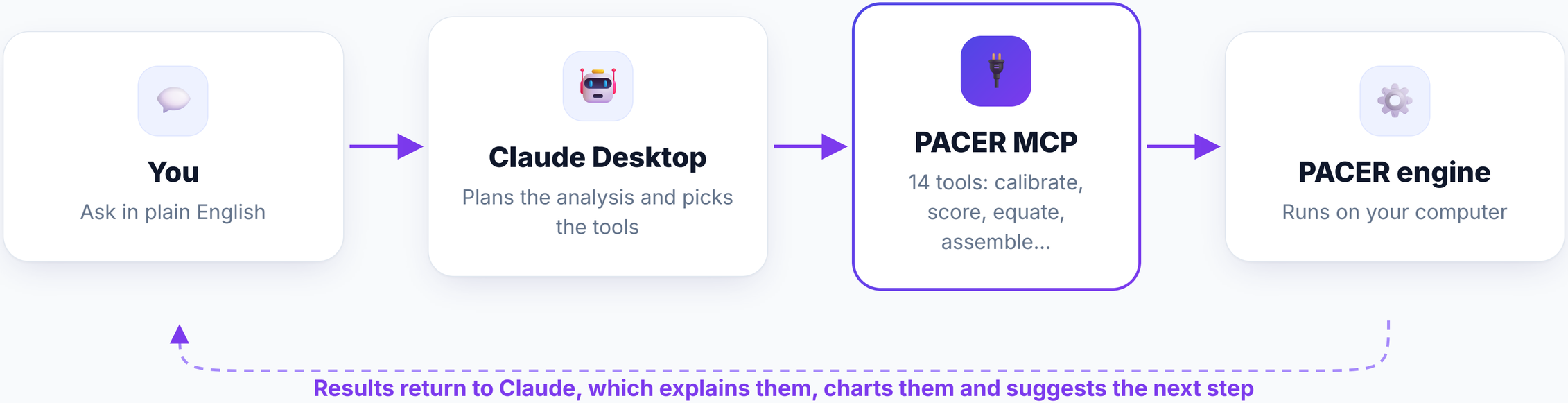
PART TWO · LIVE DEMO

PACER through Claude

The same engine and the same answers — requested in plain English and explained back to you.

HOW IT WORKS

Claude asks, PACER *computes*



🔗 You need the **PACER desktop app** plus **Claude Desktop** — or any MCP-capable desktop or command-line assistant.

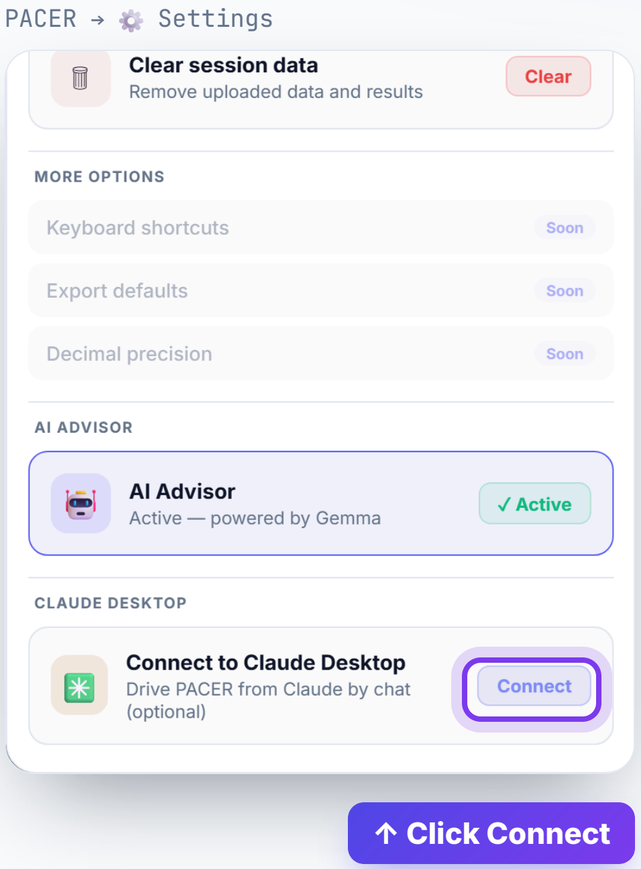
ONE-TIME SETUP

Connect Claude *in one click*

YOU NEED

The **PACER desktop app** and **Claude Desktop** — or another MCP-capable desktop or command-line assistant (setup in the appendix).

1. Click ⚙ in PACER's top bar to open **Settings**
2. Under **Claude Desktop**, click **Connect**
3. Restart **Claude Desktop** (click Connect again after each PACER upgrade)
4. Ask: `What PACER tools do you have?`



DATA PROTECTION WITH CLAUDE

Where your data *actually goes*

🖥️ YOUR COMPUTER

DATA FILES

Responses, Q-matrix, item parameters

Read from disk, never uploaded

PACER ENGINE

Every calibration and score

Computed on your processor

AI ADVISOR

Local language model

Optional; answers offline, sends nothing

CLAUDE DESKTOP

Calls PACER tools locally

Optional; receives your prompts and the results its tools return

Your prompts + tool results

Anthropic

Only if you choose Claude. Response files are not uploaded, but prompts and returned results can still contain sensitive details. Without Claude, PACER stays 100% air-gapped.

WHAT WE'LL ASK

Four prompts, *one conversation*

1 DISCOVER

```
What PACER tools do you have?
```

2 CALIBRATE

```
Calibrate a 2PL on LSAT.csv.
```

3 SCORE

```
Score the examinees with EAP and summarize the
distribution.
```

4 FULL ANALYSIS

```
Using my response file and Q-matrix, compare G-DINA
with the simpler CDMs, flag problem items, profile
the examinees, and write a short technical report.
```

TRUST, BUT VERIFY

Same engine, *same answer*

In the PACER app

ITEM 1 · a **0.4863**

ITEM 1 · b **-3.3553**

=

Through Claude

ITEM 1 · a **0.4863**

ITEM 1 · b **-3.3553**

Claude never estimates anything itself. It calls PACER's engine, so with the **same data, model and settings** the numbers match the app — and Claude then explains them.

THANK YOU

Questions?

Download PACER. Run your first calibration. See what's possible.

CONTACT

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TRY IT TODAY

https://www.paceronline.com

Desktop: github.com/hdoran/PACER_Download/releases

SCAN TO VISIT

APPENDIX · CODEX, CURSOR AND OTHER ASSISTANTS

One config, *any MCP assistant*

mcp config · Windows · PACER 3.2.0

```
{
  "mcpServers": {
    "pacer": {
      "command": "C:\\Users\\YOU\\AppData\\Local\\PACER\\app-3.2.0\\PACER.exe",
      "args": ["C:\\Users\\YOU\\AppData\\Local\\PACER\\app-3.2.0\\resources\\pacer-mcp\\pacer-mcp.mjs"],
      "env": {
        "ELECTRON_RUN_AS_NODE": "1",
        "PACER_EXE": "C:\\Users\\YOU\\AppData\\Local\\PACER\\app-3.2.0\\resources\\win-x64\\IrtEstimation.exe"
      }
    }
  }
```

OpenAI Codex
Settings → Plugins → MCPs → Add; enter the same fields

Cursor · Windsurf
`~/.cursor/mcp.json`
`…/windsurf/mcp_config.json`

VS Code · Copilot
`.vscode/mcp.json`
(key `"servers"`)

Claude Code
`claude mcp add pacer …`

Gemini CLI
`~/.gemini/settings.json`

Replace **YOU** with your Windows user name; the **app-3.2.0** folder changes with each release. On macOS the command is `/Applications/PACER.app/Contents/MacOS/PACER`. `ELECTRON_RUN_AS_NODE` makes PACER run its tool server instead of opening a window.